



Old Bridge Township School District

Curriculum & Instruction Department

Course: 7th Grade Science

Department: Science

Term: School Year

Board Approval	Committee Members	Supervisor	Notes
08/18/2026	Briana Pheifer	Jeffrey Caulfield	
	Alexandra Rivera	Jeffrey Caulfield	

MISSION STATEMENT

Recognizing that each person has an inherent worth, it is the mission of the Old Bridge Public Schools to empower each student to learn to achieve success and to become a confident, productive member of a pluralistic, global society prepared to face the challenges of the 21st Century.

BELIEFS

- Learning is a lifelong process.
- All children can learn.
- Education is a shared responsibility among the schools, the child, the family and the community.
- All students are entitled to a safe, supportive learning environment.
- Self-esteem is necessary for successful learning.
- The development of curriculum is an on-going and cyclical process.
- The district should provide a dynamic curriculum that recognizes the various needs and learning styles of our students.
- An appreciation of the arts and humanities enhances the education of the whole child.
- All students should be taught to appreciate and respect the cultural differences in our society.
- All students are worthy of educational opportunity, respect, tolerance, and fairness.
- Schools should prepare students to function successfully in society.
- All children should be encouraged to rise to their highest level of achievement.

AFFIRMATIVE ACTION STATEMENT

The Old Bridge Township Public Schools provides equal educational opportunities for all students, including equal access to all school facilities, programs, equipment, staff services, financial resources, courses, or activities, and other benefits regardless of race, color, creed, religion, sex, ancestry, national origin, social, economic or academic status or physical handicap.

PHILOSOPHY

Scientific and technological advances have proliferated and now permeate most aspects of life in the 21st century. It is increasingly important that all members of our society develop an understanding of scientific and engineering concepts and processes. Learning how to construct scientific explanations and how to design evidence-based solutions provides students with tools to think critically about personal and societal issues and needs. Students can then contribute meaningfully to decision-making processes, such as discussions about climate change, new approaches to health care, and innovative solutions to local and global problems.

The goal of science education curriculum is to produce students who have gained sufficient knowledge of the practices, crosscutting concepts, and core ideas of science and engineering to engage in public discussions on science-related issues, to be critical consumers of scientific information related to their everyday lives, and to continue to learn about science throughout their lives. They should come to appreciate that science and the current scientific understanding of the world are the result of many hundreds of years of creative human endeavor. It is especially important to note that the goals are for all students, not just those who pursue careers in science, engineering, or technology or those who continue on to higher education.

PREFACE

The Science Grade 7 curriculum is aligned with the Next Generation Science Standards. It integrates Earth Science, Life Science, and Physical Science content with science and engineering practices and cross cutting concepts to develop critical thinking and problem-solving skills in all learners. The cognitively appropriate, hands on, inquiry rich program focuses on application of content through practices that deepen science understanding, are applicable to all areas of their education, and serve as a foundation for more advanced ideas and citizenship in an increasingly complex scientific and technological world.

Life science goals

Students construct a scientific explanation based on evidence for the role of photosynthesis in the cycling of matter and flow of energy into and out of organisms. They develop a model to describe how food is rearranged through chemical reactions forming new molecules that support growth and/or release energy as this matter moves through an organism. Students analyze and interpret data to provide evidence for the effects of resource availability on organisms and populations in an ecosystem. They will construct an explanation that predicts patterns of interactions among organisms across multiple ecosystems. They develop a model to describe the cycling of matter and flow of energy among living and nonliving parts of an ecosystem, including how changes affect populations. Students evaluate competing design solutions for maintaining biodiversity and ecosystem services and determine how well designs meet the criteria and constraints of the problem. Students build on their understandings of the transfer of matter and energy as they study patterns of interactions among organisms within an ecosystem. They consider biotic and abiotic factors in an ecosystem and the effects these factors have on a population. They construct explanations for the interactions in ecosystems and the scientific, economic, political, and social justifications used in making decisions about maintaining biodiversity in ecosystems. Students demonstrate grade-appropriate proficiency in analyzing and interpreting data, using models, conducting investigations, engaging in arguments, applying engineering practices to solve a problem, and communicating information.

Students develop and use a model to describe why structural changes to genes on chromosomes may affect proteins and result in harmful, beneficial, or neutral effects to the structure and function of the organism. Students analyze and interpret data for fossil record patterns that document the existence, diversity, extinction, and change of life throughout Earth's history assuming natural laws operate today as in the past. Students use anatomical data and embryonic development to infer evolutionary relationships. Students use evidence to explain genetic variations in a population to increase some individuals' ability to survive and reproduce. Students investigate technologies used to gain organisms with desired traits, using math representations to support explanation of how natural selection changes traits in a population over time.

Earth science goals

Students understand how Earth's geosystems operate by modeling the flow of energy and cycling of matter within and among different systems. Students investigate the controlling properties of important materials and construct explanations based on the analysis of real geoscience data. The rest of this unit is broken down into three sub-ideas: Earth's large-scale systems interactions, the roles of water in Earth's surface processes, and weather and climate. Students make sense of how Earth's geosystems operate by modeling the flow of energy and cycling of matter within and among different systems. A systems approach is also important here, examining the feedback between systems as energy from the Sun is transferred between systems and circulates through the ocean and atmosphere. Students are expected to demonstrate proficiency in developing and using models and analyzing and interpreting data and conducting investigations.

Unit Goals and Pacing Guide for 7th Grade Science

Marking Period 1		Marking Period 2	
Lab Safety/Module C: Ecology and the Environment (Approx. 12 weeks)		Module D: Diversity of Living Things (Approx. 12 weeks)	
Lab Safety	Lab safety and Lab equipment (Approx. 1 block day)	Patterns of change/Homologous Structures/Natural Selection/Genetic Change and Traits	Unit section based on MS-LS4-2, MS-LS4-3, MS-LS4-4, and MS-LS3-1 (Approx. 10 block days/4 weeks)
Photosynthesis/Respiration	Unit section based on MS-LS1-6 and MS-LS1-7 (Approx. 10 block days/4 weeks)	Artificial Selection/Genetic Engineering	Unit section based on MS-LS4-5 and MS-LS4-6 (Approx. 10 block days/4 weeks)
Resource Availability in Ecosystems/Parts of an Ecosystem/Patterns of Interaction	Unit section based on MS-LS2-7, MS-LS2-1, and MS-LS2-2 (Approx. 15 block days/ 6 weeks)	Fossil Record	Unit section based on MS-LS4-1 (Approx. 10 block days/4 weeks)
Changes in an ecosystem/Bio diversity	Unit section based on MS-LS2-4 and MS-LS2-5 (Approx. 5 block days/ 2 weeks)		
Marking Period 3		Marking Period 4	
Module F: Geological Processes and History (Approx. 6 weeks)		Module E: Earth's Water and Atmosphere (Approx. 5 weeks)	
Age of Rocks/Weathering, Erosion, and Deposition/Rock Cycle/Plate Tectonics	Unit section based on MS-ESS1-4, MS-ESS2-1, MS-ESS2-2, and MS-ESS2-3 (Approx. 15 block days/ 6 weeks)	Water Cycle/Weather and Climate/Influences on Climate	Unit section based on MS-ESS2-4, MS-ESS2-5, and MS-ESS2-6 (Approx. 13 block days/5 weeks)

Core Instructional Resources and Materials (Including Varied Levels of Text)
HMH Science Dimensions Modules C, D, E, and F; Houghton Mifflin Harcourt; copyright Houghton Mifflin Harcourt Publishing Company; recommended by the committee for adoption includes eBook access for each student.

Module C: Ecology and the Environment	Duration: 30 blocks
Topic: Photosynthesis and Respiration, Resource Availability in Ecosystems, Parts of an Ecosystem, Patterns of Interaction, Changes in an Ecosystem, Biodiversity	
Alignment to NJSL Standards (include practice standards)	
- MS-LS1-6 Construct a scientific explanation based on evidence for the role of photosynthesis in the cycling of matter and flow of energy into and out of organisms - MS-LS1-7 Develop a model to describe how food is rearranged through chemical reactions forming new molecules that support growth and/or release energy as this matter moves through an organism - MS-LS2-1 Conduct an investigation to provide evidence that living things are made of cells; either one cell or many different numbers and types of cells - MS-LS2-2 Develop and use a model to describe the function of a cell as a whole and ways the parts of cells contribute to the function - MS-LS2-4 Use argument based on empirical evidence and scientific reasoning to support an explanation for how characteristic animal behaviors and specialized plant structures affect the probability of successful reproduction of animals and plants respectively - MS-LS2-5 Construct a scientific explanation based on evidence for how environmental and genetic factors influence the growth of organisms	
Technology, Career Awareness, and 21st Century Standards (NJSL 9)	
Influence of science, engineering and technology on society and the natural world. The use of technologies and any limitations on their use are driven by individual or societal needs, desires and values by the findings of scientific research; and by differences in such factors as climate; natural resources, and economic conditions. (MS-ETS1-1) New Jersey Career Readiness Practices NJCCCS 9.0 http://www.state.nj.us/education/cccs/2014/career/CareerReadyPractices.pdf CRP1. Act as a responsible and contributing citizen and employee. (MS-ESS3-1), (MS-ESS3-2), (MS-ESS3-3), (MS-ESS3-4), (MS-ESS3-5) CRP2. Apply appropriate academic and technical skills. (MS-ESS3-1), (MS-ESS3-2), (MS-ESS3-3), (MS-ESS3-4), (MS-ESS3-5) CRP5. Consider the environmental, social, and economic impacts of decisions. (MS-ESS3-1), (MS-ESS3-2), (MS-ESS3-3), (MS-ESS3-4), (MS-ESS3-5) CRP12. Work productively in teams while using cultural global competence. (MS-ESS3-1), (MS-ESS3-2), (MS-ESS3-3), (MS-ESS3-4), (MS-ESS3-5) New Jersey Technical Standards http://www.state.nj.us/education/aps/cccs/tech/ 8.1 Educational Technology (MS-ESS3-1),(MS-ESS3-2),(MS-ESS3-3),(MS-ESS3-4),(MS-ESS3-5) 8.2 Technology Education, Engineering, Design and Computational Thinking – Programming (MS-ESS3-1),(MS-ESS3-2),(MS-ESS3-3),(MS-ESS3-4),(MS-ESS3-5)	

Interdisciplinary Standards

CCSS ELA/Literacy

RST.6-8.1 Cite specific textual evidence to support analysis of science and technical texts.

RI.8.8 Trace and evaluate the argument and specific claims in a text, assessing whether the reasoning is sound and the evidence is relevant and sufficient to support the claims.

WHST.6-8.1 Write arguments to support claims with clear reasons and relevant evidence.

WHST.6-8.9 Draw evidence from informational texts to support analysis, reflection, and research.

CCSS Math

MP.4 Model with mathematics.

6.SP.A.3 Use ratio and rate reasoning to solve real-world and mathematical problems.

Curriculum Requirements (N.J.A.C. 6A:8, 6A:30)

- **Climate Change Mandate**

- Students construct an explanation connecting photosynthesis → cellular respiration → climate change. They model how plants capture CO₂ and energy from the sun (MS-LS1-6), then trace how that energy and matter move through food chains via cellular respiration (MS-LS1-7). Students analyze how increased atmospheric CO₂ from fossil fuels disrupts this cycle, reducing photosynthetic efficiency and accelerating climate change. They present evidence showing how rising temperatures affect the rate of both photosynthesis and cellular respiration in organisms, creating a feedback loop that impacts global climate.

- **Diversity, Equity, and Inclusion**

- Example: Students study photosynthesis and cellular respiration across diverse organisms and environments, not just terrestrial plants. They investigate how aquatic organisms (algae, seaweed) in ecosystems valued by Pacific Islander and Southeast Asian communities photosynthesize and transfer energy.

Learning Objectives and Activities

- **SWBAT:**

- Construct a scientific explanation based on evidence for the role of photosynthesis in the cycling of matter and flow of energy into and out of organisms.
- Develop a model to describe how food is rearranged through chemical reactions forming new molecules that support growth and/or release energy as this matter moves through an organism.
- Explain how producers produce their own food
- Analyze and interpret data to provide evidence for the effects of resource availability on organisms and populations of organisms in an ecosystem.
- Construct an explanation that predicts patterns of interactions among organisms across multiple ecosystems.
- Construct an argument supported by empirical evidence that changes to physical or biological components of an ecosystem affect populations.
- Evaluate competing design solutions for maintaining biodiversity and ecosystem services.

- SWBAT demonstrate understanding of the following:
 - Plants, algae, and many microorganisms use energy from light to make sugars from carbon dioxide and water through photosynthesis which releases oxygen. These sugars can be used immediately or stored for later use.
 - Within individual organisms, food moves through a series of chemical reactions in which its broken down and rearranged to form new molecules, to support growth, or to release energy.
 - Organisms, and populations of organisms are dependent on their environmental interactions both with other living things and with non-living factors.
 - In ecosystems, organisms and populations with similar requirements for food, water, oxygen or other resources may compete for limited access to these resources, which restrains their growth and reproduction.
 - Growth of organisms and populations are limited by access to resources.
 - Predatory interactions may reduce the number of organisms or eliminate whole populations of organisms. Mutually beneficial interactions, in contrast, may become so interdependent that each organism requires the other for survival. Although species involved in these competitive, predatory, and mutually beneficial interactions vary across ecosystems, the patterns of interactions of organisms with their environments, both living, and nonliving are shared.
 - Food webs are models that demonstrate how matter and energy is transferred between producers, consumers and decomposers as the three groups interact within an ecosystem. Transfers of matter occur into and out of the physical environment at every level. Decomposers recycle nutrients from dead plants or animals back to the soil in a terrestrial environment or water in an aquatic environment. The atoms that make up the organisms are cycled repeatedly between the living and non-living parts of the ecosystem.
 - Ecosystems are dynamic in nature; their characteristics can vary over time. Disruptions to any physical or biological component of an ecosystem can lead to shifts in all its populations. Biodiversity describes the variety of species found in Earth's terrestrial and oceanic ecosystems. The completeness or integrity of an ecosystem's biodiversity is often used as a measure of its health.

Learning Activities:

- Content Knowledge resources:
 - HMH Science Dimensions Module C Unit 1; Lessons 1-3
 - HMH Science Dimensions Module C Unit 2; Lessons 1-3
- Activity resources:
 - Energy in Organisms and Photosynthesis PPT and notes.
 - Cellular Respiration PPT and notes.
 - Energy in food webs PPT and notes.
 - Parts of ecosystem PPT and notes.
 - Patterns of Interactions PPT and notes.
- Assessment resources:
 - Energy in Organisms and Photosynthesis apply knowledge questions.

Benchmark, Formative, Summative and Alternative Assessments

- Alternative
 - Kahoot
 - Blooket
 - Wayground
 - Nearpod
- Summative
 - Photosynthesis Quiz
 - Photosynthesis & Cellular Respiration Unit Test
 - Energy Food Pyramid Lab
 - Ecosystem Levels Project- Food Web
 - Ecology Test
 - Kelp Forest Project
- Formative
 - Classwork
 - Teacher Observation

Module D: Diversity of Living Things	Duration: 30 blocks
Topic: Patterns of Change, Homologous Structures, Natural Selection, Genetic Change and Traits, Artificial Selection, Genetic Engineering & Fossil Record	
Alignment to NJSLs (include practice standards)	
● MS-LS4-1 Analyze and interpret data for patterns in the fossil record that document the existence, diversity, extinction, and change of life forms throughout the history of life on Earth under the assumption that natural laws operate today as in the past ● MS-LS4-2 Apply scientific ideas to construct an explanation for the anatomical similarities and differences among modern organisms and between modern and fossil organisms to infer evolutionary relationships ● MS-LS4-3 Analyze displays of pictorial data to compare patterns of similarities in the embryological development across multiple species to identify relationships not evident in the fully formed anatomy ● MS-LS4-4 Construct an explanation based on evidence that describes how genetic variations of traits in a population increase some individuals' probability of surviving and reproducing in a specific environment ● MS-LS4-5 Gather and synthesize information about technologies that have changed the way humans influence the inheritance of desired traits in organisms ● MS-LS4-6 Use mathematical representations to support explanations of how natural selection may lead to increases and decreases of specific traits in populations over time	
Technology, Career Awareness, and 21st Century Standards (NJSLs 9)	
Influence of science, engineering and technology on society and the natural world. The use of technologies and any limitations on their use are driven by individual or societal needs, desires and values by the findings of scientific research; and by differences in such factors as climate; natural resources, and economic conditions. (MS-ETS1-1)	

New Jersey Career Readiness Practices

NJCCCS 9.0 <http://www.state.nj.us/education/cccs/2014/career/CareerReadyPractices.pdf>

CRP1. Act as a responsible and contributing citizen and employee. (MS-ESS3-1), (MS-ESS3-2), (MS-ESS3-3), (MS-ESS3-4), (MS-ESS3-5)

CRP2. Apply appropriate academic and technical skills. (MS-ESS3-1), (MS-ESS3-2), (MS-ESS3-3), (MS-ESS3-4), (MS-ESS3-5)

CRP5. Consider the environmental, social, and economic impacts of decisions. (MS-ESS3-1), (MS-ESS3-2), (MS-ESS3-3), (MS-ESS3-4), (MS-ESS3-5)

CRP12. Work productively in teams while using cultural global competence. (MS-ESS3-1), (MS-ESS3-2), (MS-ESS3-3), (MS-ESS3-4), (MS-ESS3-5)

New Jersey Technical Standards

<http://www.state.nj.us/education/aps/cccs/tech/>

8.1 Educational Technology (MS-ESS3-1),(MS-ESS3-2),(MS-ESS3-3),(MS-ESS3-4),(MS-ESS3-5)

8.2 Technology Education, Engineering, Design and Computational Thinking – Programming (MS-ESS3-1),(MS-ESS3-2),(MS-ESS3-3),(MS-ESS3-4),(MS-ESS3-5)

Interdisciplinary Standards

CCSS ELA/Literacy

RST.6-8.1 Cite specific textual evidence to support analysis of science and technical texts.

RST.6-8.4 Determine the meaning of symbols, key terms and other domain-specific words and phrases as they are used in a specific scientific or technical context relevant to grades 6-8 texts and topics.

RST.6-8.7 Integrate quantitative or technical information expressed in words in a text with a version of that information expressed visually (flow chart, diagram, graph, table etc.)

SL.8.5 Include multimedia components and visual displays in presentations to clarify claim and findings and emphasize salient points.

RST.6-8.1 Cite specific textual evidence to support analysis of science and technical texts.

CCSS Math

MP.4 Model with mathematics.

Curriculum Requirements (N.J.A.C. 6A:8, 6A:30)

● **Diversity, Equity, and Inclusion**

- Example: Investigate how mutations in rice varieties developed by Asian farmers created drought-resistant, disease-resistant, or nutrient-rich strains (like Golden Rice). Students model how these mutations affected protein production and organism function, recognizing diverse agricultural contributions to genetic innovation.

Learning Objectives and Activities

● **SWBAT:**

- Explain how fossil data can be used to provide evidence of the history of life on Earth.
- Explore ways fossils can form
- Describe how fossil record is constructed

● **SWBAT demonstrate understanding of the following:**

- Anatomical similarities and differences between various organisms living today and between them and organisms in the fossil record, enable the reconstruction of evolutionary history and the inference of lines of evolutionary descent.
- Comparison of the embryological development of different species also reveals similarities that show relationships not evident in the fully formed anatomy.
- Genes are located in the chromosomes of cells, with each chromosome pair containing two variants of each of many distinct genes. Each distinct gene chiefly controls the production of specific proteins, which in turn affects the individual's traits. Changes (mutations) to genes can result in changes to proteins, which can affect the structures and functions of the organism and thereby change traits.
- In addition to variations that arise from sexual reproduction, genetic information can be altered because of mutations. Though rare, mutations may result in changes to the structure and function of proteins. Some changes are beneficial, others harmful, and some neutral to the organism.
- In artificial selection, humans have the capacity to influence certain characteristics of organisms by selective breeding. One can choose desired parental traits determined by genes, which are then passed on to offspring.
- Adaptation by natural selection acting over generations is one important process by which species change over time in responses to changes in environmental conditions. Traits that support successful survival and reproduction in the new environment become more common; those that do not become less common. Thus, the distribution of traits in a population change.
- Natural selection leads to the predominance of certain traits in a population, and the suppression of others.
- The collection of fossils and their placement in chronological order (e.g., through the location of the sedimentary layers in which they are found or through radioactive dating) is known as the fossil record. It documents the existence, diversity, extinction, and change of many life forms throughout the history of life on Earth.

Learning Activities:

- Content Knowledge resources:
 - HMH Science Dimensions Module D
- Activity resources:
 - Biodiversity WebQuest
 - Role of bees in pollination worksheet
 - Amazon rainforest case study video and questions.
 - Evidence of Common Ancestry Questions pg. 6-13
 - Evolution PPT and notes
 - Evolution of the horse
 - Charles Darwin BrainPOP
 - Artificial selection and genetic engineering PPT and notes
 - Animal Adaptations project
 - Fossils PPT notes
 - Inferring the Age of Fossils activity
 - Getting into the Fossil Record Form website and questions
- Assessment resources:
 - Biodiversity quiz
 - Evolution test
 - Natural/Artificial Selection & Genetic Engineering Test
 - Animal Adaptations project
 - Fossils quiz

Benchmark, Formative, Summative and Alternative Assessments

- Alternative
 - Kahoot
 - Blooket
 - Wayground
 - Nearpod
- Summative
 - Biodiversity quiz
 - Evolution test
 - Natural/Artificial Selection & Genetic Engineering Test
 - Animal Adaptations project
 - Fossils quiz
- Formative
 - Classwork
 - Teacher Observation

Module F: Geological Processes and History	Duration: 15 blocks
Topic: Age of Rocks, Weathering, Erosion, and Deposition, Rock Cycle, Plate Tectonics	
Alignment to NJSLS (include practice standards)	
● MS-ESS2-1 Develop and use a model of the Earth-sun-moon system to describe the cyclic patterns of lunar phases, eclipses of the sun and moon, and seasons. ● MS-ESS2-2 Develop and use a model to describe the role of gravity in the motions within galaxies and the solar system ● MS-ESS2-3 Analyze and interpret data to determine scale properties of objects in the solar system.	
Technology, Career Awareness, and 21st Century Standards (NJSLS 9)	
New Jersey Career Readiness Practices NJCCCS 9.0 http://www.state.nj.us/education/cccs/2014/career/CareerReadyPractices.pdf CRP1. Act as a responsible and contributing citizen and employee. (MS-ESS3-1),(MS-ESS3-2),(MS-ESS3-3),(MS-ESS3-4),(MS-ESS3-5) CRP2. Apply appropriate academic and technical skills.(MS-ESS3-1),(MS-ESS3-2),(MS-ESS3-3),(MS-ESS3-4),(MS-ESS3-5) CRP5. Consider the environmental, social and economic impacts of decisions. (MS-ESS3-1),(MS-ESS3-2),(MS-ESS3-3),(MS-ESS3-4),(MS-ESS3-5) CRP12. Work productively in teams while using cultural global competence. (MS-ESS3-1),(MS-ESS3-2),(MS-ESS3-3),(MS-ESS3-4),(MS-ESS3-5) New Jersey Technical Standards http://www.state.nj.us/education/aps/cccs/tech/ 8.1 Educational Technology (MS-ESS3-1),(MS-ESS3-2),(MS-ESS3-3),(MS-ESS3-4),(MS-ESS3-5)	

8.2 Technology Education, Engineering, Design and Computational Thinking – Programming (MS-ESS3-1),(MS-ESS3-2),(MS-ESS3-3),(MS-ESS3-4),(MS-ESS3-5)

Interdisciplinary Standards

CCSS ELA/Literacy

SL.8.5 Include multimedia components and visual displays in presentations to clarify claims and findings and emphasize salient points.

RST.6-8.1 Cite specific textual evidence to support analysis of science and technical texts.

WHST.6-8.2 Write informative/explanatory texts to examine a topic and convey ideas, concepts and information through the selection, organization and analysis of relevant content.

SL.8.5 Include multimedia components and visual displays in presentations to clarify claims and findings and emphasize salient points.

RST.6-8.1 Cite specific textual evidence to support analysis of science and technical texts.

CCSS Math

MP.2 Reason abstractly and quantitatively.

6.EE.B.6 Use variables to represent numbers and write expressions when solving a real-world or mathematical problem; understand that a variable can represent an unknown number or depending on the purpose at hand, any number in a specified set.

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MP.2 Reason abstractly and quantitatively.

Curriculum Requirements (N.J.A.C. 6A:8, 6A:30)

- **History of Disabled and LGBTQ+ Persons**

- Students learn about LGBTQ+ Earth scientists and geochemists who advanced our understanding of Earth's systems. Example: Research pioneering LGBTQ+ scientists who discovered biogeochemical cycles or developed models of Earth processes. Students develop their own models using multiple modalities and accessible tools—digital simulations, physical manipulatives, kinesthetic demonstrations, tactile models—recognizing that diverse ways of thinking and learning strengthen scientific understanding. Highlight how blind or visually impaired geoscientists have contributed using audio descriptions, 3D models, and collaborative approaches.

- **AAPI Mandate**

- Students research AAPI Earth scientists and their contributions to understanding material and energy cycling. Example: Investigate how Chinese hydrologists study the monsoon system and water cycling in Asia, or how Japanese volcanologists study geochemical cycling through volcanic activity. Research AAPI traditional knowledge about seasonal cycles, water management, and sustainable resource use. Students develop models showing how the water cycle, carbon cycle, and energy flow operate in Asian ecosystems (monsoon regions, Himalayan systems, Pacific Island systems). They incorporate AAPI scientific discoveries and traditional ecological knowledge into their models, recognizing that understanding Earth's cycles has deep cultural and scientific roots in Asian communities.

Learning Objectives and Activities

- SWBAT:
 - Construct a scientific explanation based on evidence from rock strata for how the geologic time scale is used to organize Earth's 4.6-billion-year-old history.
 - Develop a model to describe the cycling of Earth's materials and the flow of energy that drives this process.
 - Construct an explanation based on evidence for how geoscience processes have changed Earth's surface at varying time and spatial scales.
 - Analyze and interpret data on the distribution of fossils and rocks, continental shapes, and seafloor structures to provide evidence of the past plate motions.
- SWBAT demonstrate understanding of the following:
 - The geologic time scale interpreted from rock strata provides a way to organize Earth's history. Analysis of rock strata and the fossil record provide only relative dates, not an absolute scale.
 - All Earth's processes are the result of energy flowing and matter cycling within and among the planet's systems. This energy is derived from the sun and Earth's hot interior. The energy that flows and matters that cycles produce chemical and physical changes in Earth's materials and living organisms.
 - The planet's systems interact over scales that range from microscopic to global in size; and they operate over fractions of a second to billions of years. These interactions have shaped Earth's history and will determine its future.
 - Water movements - both on the land and underground - cause weathering and erosion, which change the land's surface features and create underground formations.
 - Tectonic processes continually generate new ocean sea floors at ridges and destroy old sea floors at trenches.
 - Maps of ancient land and water patterns, based on investigations of rocks and fossils, make clear how Earth's plates have moved great distances, collided and spread apart.

Learning Activities:

- Content Knowledge resources:
 - HMH Science Dimensions Module F
- Activity resources:
 - Geologic Time Scale WebQuest
 - Geologic Time Scale BrainPOP
 - Weathering, Erosion, and Deposition PPT and notes
 - Weathering/Erosion study jams video with questions
 - Types of Weathering video and questions
 - Weathering BrainPOP
 - Bill Nye- Weathering, Erosion, and Deposition
 - Igneous, Sedimentary, and Metamorphic Rock PPT and notes
 - Rocks and Minerals Video Lab
 - Rock Cycle diagram
 - Starburst video lab and questions
 - Continental drift PPT
 - Seafloor spreading PPT
 - Plate boundaries PPT notes

○ Convection currents/energy flow PPT notes ● Assessment resources: ○ Geologic Processes and History apply knowledge of questions. ○ Weathering/erosion quiz ○ Rocks test ○ Seafloor spreading
Benchmark, Formative, Summative and Alternative Assessments
● Alternative ○ Kahoot ○ Blooket ○ Wayground ○ Nearpod ● Summative ○ Geologic Timescale Postcard Project ○ Weathering/erosion Quiz ○ Weathering, Erosion, and Deposition WebQuest ○ Test- Rocks ○ Test- Seafloor spreading ● Formative ○ Classwork ○ Teacher Observation

Module E: Weather and Climate	Duration: 13 blocks
Topic: Water Cycle, Weather and Climate, & Influences on Climate	
Alignment to NJSLs (include practice standards)	
● MS-ESS2-4 Develop a model to describe the cycling of water through Earth's systems driven by energy from the sun and the force of gravity ● MS-ESS2-5 Collect data to provide evidence for how the motions and complex interactions of air masses result in changes in weather conditions ● MS-ESS2-6 Develop and use a model to describe how unequal heating and rotation of the Earth cause patterns of atmospheric and oceanic circulation that determine regional climates	
Technology, Career Awareness, and 21st Century Standards (NJSLs 9)	
Influence of science, engineering and technology on society and the natural world.	
New Jersey Career Readiness Practices	

NJCCCS 9.0 <http://www.state.nj.us/education/cccs/2014/career/CareerReadyPractices.pdf>

CRP1. Act as a responsible and contributing citizen and employee. (MS-ESS3-1), (MS-ESS3-2), (MS-ESS3-3), (MS-ESS3-4), (MS-ESS3-5)

CRP2. Apply appropriate academic and technical skills. (MS-ESS3-1), (MS-ESS3-2), (MS-ESS3-3), (MS-ESS3-4), (MS-ESS3-5)

CRP5. Consider the environmental, social, and economic impacts of decisions. (MS-ESS3-1), (MS-ESS3-2), (MS-ESS3-3), (MS-ESS3-4), (MS-ESS3-5)

CRP12. Work productively in teams while using cultural global competence. (MS-ESS3-1), (MS-ESS3-2), (MS-ESS3-3), (MS-ESS3-4), (MS-ESS3-5)

New Jersey Technical Standards

<http://www.state.nj.us/education/aps/cccs/tech/>

8.1 Educational Technology (MS-ESS3-1),(MS-ESS3-2),(MS-ESS3-3),(MS-ESS3-4),(MS-ESS3-5)

8.2 Technology Education, Engineering, Design and Computational Thinking – Programming (MS-ESS3-1),(MS-ESS3-2),(MS-ESS3-3),(MS-ESS3-4),(MS-ESS3-5)

Interdisciplinary Standards

CCSS ELA/Literacy

RST.6-8.1 Cite specific textual evidence to support analysis of science and technical texts.

RST.6-8.9 Compare and contrast the information gained from experiments, simulations, video or multimedia sources that gained from reading a text on the same topic.

WHST.6-8.8 Gather relevant information from multiple print and digital sources; assess the credibility of each source; and quote or paraphrase the data and conclusions of others while avoiding plagiarism and providing basic bibliographic information for sources.

SL.8.5 Include multimedia components and visual displays in presentations to clarify claims and findings and emphasize salient points.

CCSS Math

MP.2 Reason abstractly and quantitatively.

Curriculum Requirements (N.J.A.C. 6A:8, 6A:30)

- **Climate Change Mandate**

- Example: Students gather historical weather data (temperature, precipitation, wind patterns) from local and regional sources over decades. They analyze trends showing how shifting air mass patterns are creating more extreme weather events (stronger storms, prolonged droughts, unusual temperature swings). Students use evidence to construct explanations for how rising global temperatures are destabilizing atmospheric circulation patterns and changing traditional weather predictions. They connect their local data to global climate change, demonstrating how energy imbalances drive altered air mass behavior.

Learning Objectives and Activities

- **SWBAT:**

- Model air circulation in Earth's atmosphere
- Explore how air pressure systems influence weather
- Develop and use a model to describe how unequal heating and rotation of the Earth cause patterns of atmospheric and oceanic circulation that determine regional climates

- **SWBAT demonstrate understanding of the following:**

- Develop a model to describe the cycle of water through Earth's systems driven by energy from the sun and the force of gravity.

- Collect data to provide evidence for how the motions and complex interactions of air masses result in changes in weather conditions.
- Develop and use a model to describe how unequal heating and rotation of the Earth cause patterns of atmospheric and oceanic circulation that determine regional climates.

Learning Activities:

- Content Knowledge resources:
 - HMH Science Dimensions Module E
- Activity resources:
 - Water Cycle BrainPOP
 - Winds PPT notes
 - Wind currents and Weather patterns Video and questions
 - What are Convection currents Mr. Wizard's Supermarket Science video
 - Earth's Atmosphere BrainPOP
 - Circulation in Earth's Oceans PPT notes
 - Wind and Wind Currents WebQuest
 - Water in the Atmosphere PPT notes
 - Air Masses and Fronts PPT notes
 - Live demo page 88 HMH book
- Assessment resources:
 - Water Cycle Diagram
 - Circulation in Earth's Atmosphere Test
 - Create a Weather Map Project

Benchmark, Formative, Summative and Alternative Assessments

- Alternative
 - Kahoot
 - Blooket
 - Wayground
 - Nearpod
- Summative
 - Circulation in Earth's Atmosphere Test
 - Create a Weather Map Project
- Formative
 - Classwork
 - Teacher Observation

Modifications (ML, Special Education, At Risk Students, Gifted & Talented, & 504 Plans)

ML

- Work toward longer passages as skills in English increase

- Use visuals
- Introduce key vocabulary before lesson
- Teacher models reading aloud daily
- Use of Bilingual Dictionary
- Guided notes and/or scaffold outline for written assignments

Supports for Students With IEPs:

- Allow extra time to complete assignments or tests
- Guided notes and/or scaffold outline for written assignments
- Work in a small group
- Allow answers to be given orally or dictated
- Use large print books, Braille, or e-books
- Follow all IEP modifications

At-Risk Students:

- Guided notes and/or scaffold outline for written assignments
- Introduce key vocabulary before lesson
- Work in a small group
- Lesson taught again using a differentiated approach
- Allow answers to be given orally or dictated
- Use visuals / anchor charts
- Leveled texts according to ability

Gifted and Talented:

- Create an enhanced set of introductory activities (e.g. advance organizers, concept maps, concept puzzles)
- Provide options, alternatives and choices to differentiate and broaden the curriculum
- Organize and offer flexible small group learning activities
- Provide whole group enrichment explorations
- Use center, stations, or contracts
- Organize integrated problem-solving simulations
- Propose interest-based extension activities
- Expose students to beyond level texts.

Supports for Students With 504 Plans:

- Follow all the 504 plan modifications
- Text to speech/audio recorded selections
- Amplification system as needed
- Leveled texts according to ability
- Fine motor skill stations embedded in rotation as needed
- Provide anchor charts with high frequency words and phonemic patterns